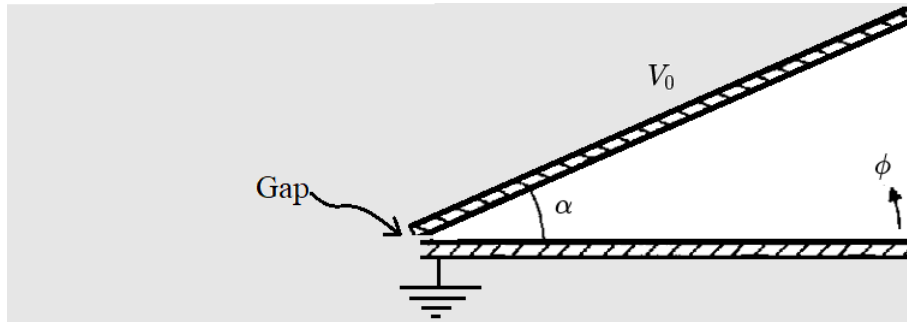




Q1 [25]: Two infinite insulated conducting planes maintained at potentials 0 and V_0 form a wedge-shaped configuration, as shown in Fig. 1. Determine the potential distribution and electric field for the region $\alpha < \phi < 2\pi$.



Solution: Here we have $\partial V/\partial r = 0$ and $\partial V/\partial z = 0$. Since the region at $r = 0$ is excluded, Laplace's equation becomes

$$\frac{d^2 V}{d\phi^2} = 0$$

V depends only on ϕ , and can be obtained from above Eq. by integrating twice.

$$V(\phi) = K_1 \phi + K_2$$

The two integration constants K_1 and K_2 are to be determined from boundary values. For $\alpha \leq \phi \leq 2\pi$:

$$\text{At } \phi = \alpha, \quad V(\alpha) = \alpha K_1 + K_2 = V_0. \quad \text{--- (a)}$$

$$\text{At } \phi = 2\pi, \quad V(2\pi) = 2\pi K_1 + K_2 = 0. \quad \text{--- (b)}$$

Solving (a) and (b), we find

$$K_1 = -\frac{V_0}{2\pi - \alpha}, \quad K_2 = \frac{2\pi V_0}{2\pi - \alpha}.$$

Finally, we obtain

$$\boxed{\begin{aligned} V(\phi) &= V_0 \frac{2\pi - \phi}{2\pi - \alpha}, \quad \alpha \leq \phi \leq 2\pi. \\ \mathbf{E} &= -\frac{1}{r} \frac{\partial V}{\partial \phi} \hat{\phi} = \frac{V_0}{r} \frac{\hat{\phi}}{2\pi - \alpha}. \end{aligned}}$$

Q2 [50]: The axis of a dielectric tube of inner radius r_i and outer radius r_0 coincides with z -axis. A polarization vector $\mathbf{P} = P_0 (3x \hat{\mathbf{x}} + 5y \hat{\mathbf{y}})$ exists inside.

- (a) Determine the polarization surface charge density at $r = r_i$ and $r = r_0$.
- (b) Determine the polarization volume charge density for $r_i < r < r_0$.
- (c) Find bound charges at $r = r_i$ and $r = r_0$ surfaces (for a unit length).
- (d) Find bound charges in the region $r_i < r < r_0$ (for a unit length).
- (e) Is the (per-unit-length) total bound charge equal to zero?

Solution:

(a) Surface charge density at outer surface ($r = r_0$)

The normal vector on the outer surface is in the outward direction!!!

$$\begin{aligned}\rho_{ps}(\text{outer}) &= \mathbf{P} \cdot \hat{\mathbf{n}} = \mathbf{P} \cdot \hat{\mathbf{r}} = P_0 (\hat{\mathbf{x}} 3x + \hat{\mathbf{y}} 5y) \cdot \hat{\mathbf{r}} \\ &= P_0 (\hat{\mathbf{x}} 3r_0 \cos \phi + \hat{\mathbf{y}} 5r_0 \sin \phi) \cdot (\hat{\mathbf{x}} \cos \phi + \hat{\mathbf{y}} \sin \phi) \\ &= P_0 r_0 (3 \cos^2 \phi + 5 \sin^2 \phi) = P_0 r_0 (3 + 2 \sin^2 \phi) .\end{aligned}$$

Surface charge density at inner surface ($r = r_i$)

The normal vector on the inner surface is in the inward direction!!!

$$\begin{aligned}\rho_{ps}(\text{inner}) &= \mathbf{P} \cdot \hat{\mathbf{n}} = \mathbf{P} \cdot (-\hat{\mathbf{r}}) = -P_0 (\hat{\mathbf{x}} 3x + \hat{\mathbf{y}} 5y) \cdot \hat{\mathbf{r}} \\ &= -P_0 (\hat{\mathbf{x}} 3r_i \cos \phi + \hat{\mathbf{y}} 5r_i \sin \phi) \cdot (\hat{\mathbf{x}} \cos \phi + \hat{\mathbf{y}} \sin \phi) \\ &= -P_0 r_i (3 \cos^2 \phi + 5 \sin^2 \phi) = -P_0 r_i (3 + 2 \sin^2 \phi) .\end{aligned}$$

(b) Volumetric charge density

$$\mathbf{P} = P_0 (\hat{\mathbf{x}} 3x + \hat{\mathbf{y}} 5y). \Rightarrow \rho_{pv}(\text{inside}) = -\nabla \cdot \mathbf{P} = -8P_0.$$

(c) Bound charges at the outer surface

$$Q_{ps}(\text{outer}) = \int_{\phi=0}^{2\pi} \rho_{ps}(\text{outer}) r_0 d\phi = P_0 r_0^2 \int_{\phi=0}^{2\pi} (3 + 2 \sin^2 \phi) d\phi = 8\pi P_0 r_0^2.$$

Bound charges at the inner surface

$$Q_{ps}(\text{inner}) = \int_{\phi=0}^{2\pi} \rho_{ps}(\text{inner}) r_i d\phi = -P_0 r_i^2 \int_{\phi=0}^{2\pi} (3 + 2 \sin^2 \phi) d\phi = -8\pi P_0 r_i^2.$$

$$\text{Bound charges inside } Q_{pv}(\text{inside}) = \int_v \rho_{pv}(\text{inside}) dv = -8P_0(\pi(r_0^2 - r_i^2)).$$

$$\text{Total charge} = Q_{pv}(\text{inside}) + Q_{ps}(\text{inner}) + Q_{ps}(\text{outer}) = 0.$$

Q3 [25]: Assume that the $z = 0$ plane separates two dielectric regions with $\epsilon_{r1} = 2$ and $\epsilon_{r2} = 3$. If we know that $\mathbf{E}_1$ in region 1 is $2y\hat{x} - 3x\hat{y} + (5+z)\hat{z}$, what do we also know about $\mathbf{E}_2$ and $\mathbf{D}_2$ in region 2 (in the vicinity of boundary)?

Solution:

At the $z=0$ plane:

$$\bar{\mathbf{E}}_1 = 2y\hat{x} - 3x\hat{y} + 5\hat{z} \Big|_{z=0}$$

$$\bar{\mathbf{E}}_{1t} = \bar{\mathbf{E}}_{2t} = 2y\hat{x} - 3x\hat{y} \Big|_{z=0}$$

Since, no deliberate surface charge density is mentioned,

Therefore

$$\bar{\mathbf{D}}_{1n} = \bar{\mathbf{D}}_{2n} \Rightarrow \epsilon_1 \bar{\mathbf{E}}_{1n} = \epsilon_2 \bar{\mathbf{E}}_{2n} \text{ (at } z=0\text{)}$$

$$2\bar{\mathbf{E}}_{1n} = 3\bar{\mathbf{E}}_{2n}$$

$$\Rightarrow \bar{\mathbf{E}}_{2n} = \frac{2}{3}\bar{\mathbf{E}}_{1n} = \frac{10}{3}\hat{z}$$

$$\therefore \bar{\mathbf{E}}_2 = 2y\hat{x} - 3x\hat{y} + \frac{10}{3}\hat{z} \text{ (at } z=0\text{)}$$

$$\Rightarrow \bar{\mathbf{D}}_2 = \epsilon_2 \bar{\mathbf{E}}_2 = 3\epsilon_0 \left(2y\hat{x} - 3x\hat{y} + \frac{10}{3}\hat{z} \right).$$